

USED MEDIAS LLC

EMBEDDED SYSTEMS DIVISION

Engineering Documentation

cyd-os

DOCUMENT	REVISION	STATUS	CLASSIFICATION
cyd-os	—	—	Internal

Purpose of this set

These reports record the design of cyd-os in enough detail that the project can be picked up cold — by someone else, or by the author after six months away — without re-deriving decisions from the source.

Each report states **what was decided, what it was measured against, and what remains unverified**. Claims that were tested on hardware are marked as such. Claims taken from documentation or reasoning are marked separately, because on a from-scratch kernel the difference between "this is true" and "this should be true" is the difference between a working boot and a silent reboot.

Index

ID	Title	Covers
UM-CYDOS-001	System Architecture and Scope	Layer model, what is built vs borrowed, the isolation problem, why a bytecode VM
UM-CYDOS-002	Boot Chain and Image Format	ROM → 2nd stage → kernel, image header, measured boot trace
UM-CYDOS-003	Xtensa ABI Selection	Windowed vs call0, codegen evidence, consequences for the scheduler
UM-CYDOS-004	Memory Map and Allocation Policy	ESP32 regions, our layout, the bootloader overlap risk
UM-CYDOS-005	Build and Flash Pipeline	Toolchain, flags and why each one, reproducing a build, why not PlatformIO
UM-CYDOS-006	Milestone 0 Verification Report	Test method, captured output, pass/fail per assertion
UM-CYDOS-007	Development Roadmap M1–M5	Each milestone, its risks, and its exit criteria
UM-CYDOS-008	Milestone 1 Verification Report	Vectors, timer source and level choice, interrupt entry/exit, measured results

Reading order

New to the project: **001** → **002** → **004** → **003**. That gives the shape of the system, how it starts, where things live, and why the calling convention is unusual.

Picking up implementation work: **006** (what is known-good) then **007** (what is next and what it will break).

Reproducing a build: **005** alone is sufficient.

Status summary as of this revision

Item	State
Milestone 0 — kernel boots, self-checks pass	Complete, verified on hardware
Milestone 1 — timer interrupt, tick counter	Complete, verified on hardware
Milestone 2 — native task switching	Not started
Version control	Initialised 2026-08-14
JTAG debug probe	Ordered, not in hand
Bootloader IRAM overlap	Closed — investigated and disproved, UM-CYDOS-004 §5

Carried risk — panic handler unexercised. The fault reporting path added in M1 links and is placed correctly, but no exception has been deliberately triggered to confirm it produces output under real fault conditions. M2 is when it will first be needed in anger. Testing it deliberately before then is cheap; discovering it is broken while debugging a context switch is not.